

Architecture Design Report

This document presents the architectural design of the system, including its context, logical structure, runtime b

1. System Requirements

System Name: LibraryManagementSystem

Architecture Style: Microservices Architecture

Functional Requirements

- User registration and authentication
- Manage library members
- Catalog books and authors
- Search and browse books
- Borrow and return books
- Track borrowing history
- Send due date notifications

Non-Functional Requirements

- Scalability: Medium
- Availability: High
- Latency: Medium
- Reliability: High
- Consistency: Strong
- Security: Medium

2. Formal Architecture Specification (ACME ADL)

The following section provides a formal architectural specification using ACME ADL, capturing components, co

```
System LibraryManagementSystem {
  Property view : String = "runtime";
  Property architectureStyle : String = "Microservices Architecture";
  Property source : String = "AI";
  Component UserService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Handle user registration and authentication";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
  }
  Component MemberService = new Component {
    Port in;
```

```

    Port out;
    Property responsibility : String = "Manage library members";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component CatalogService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Catalog books and authors";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component SearchService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Search and browse books";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component BorrowingService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Borrow and return books";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component HistoryService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Track borrowing history";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Component NotificationService = new Component {
    Port in;
    Port out;
    Property responsibility : String = "Send due date notifications";
    Property state : String = "stateless";
    Property scaling : String = "horizontal";
}
Attachment UserService.out to MemberService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment UserService.out to CatalogService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment MemberService.out to CatalogService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment CatalogService.out to SearchService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;

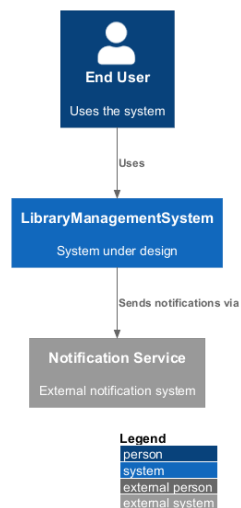
```

```

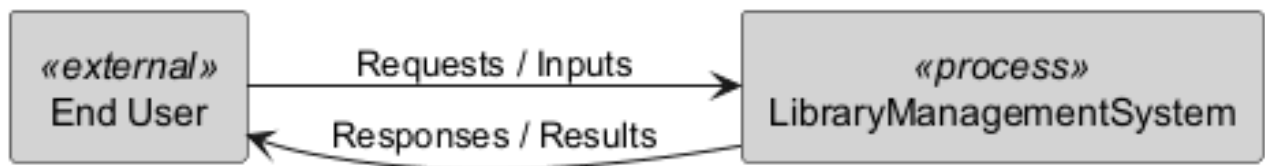
    Property backpressure : String = "bounded";
}
Attachment SearchService.out to BorrowingService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment BorrowingService.out to HistoryService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Attachment BorrowingService.out to NotificationService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = false;
    Property backpressure : String = "bounded";
}
Attachment HistoryService.out to NotificationService.in {
    Property delivery : String = "at-least-once";
    Property ordering : Boolean = true;
    Property backpressure : String = "bounded";
}
Property resilience_pattern : String = "Retry, CircuitBreaker";
Property delivery : String = "at-least-once";
Property ordering : Boolean = false;
Property authentication : String = "service-to-service";
Property authorization : String = "RBAC";
}

```

3. Context View



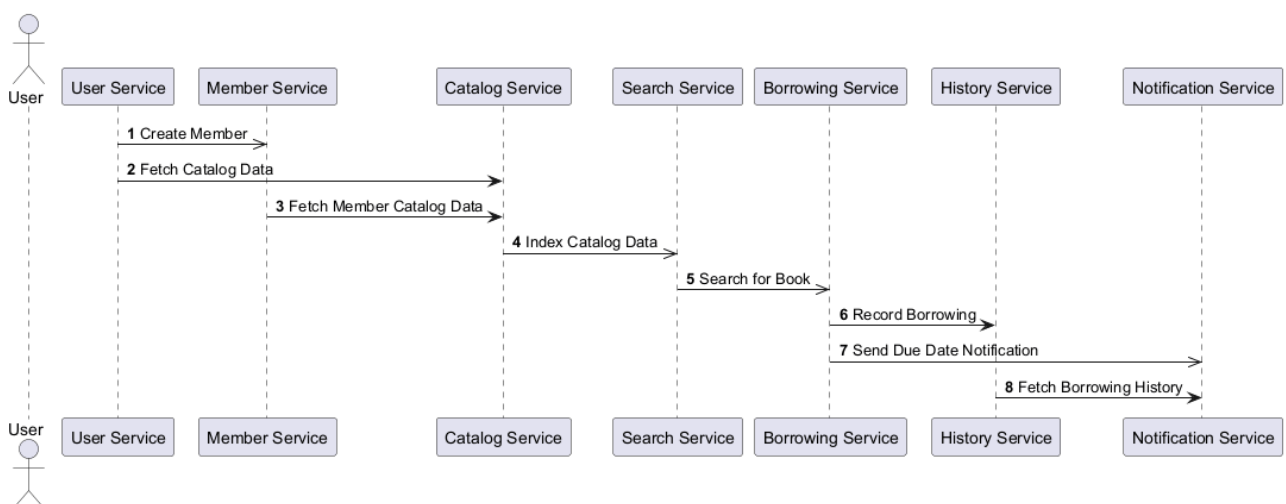
3.1 Data Context View (Level 0 DFD)



4. Logical View (C4 Container Diagram)



5. Process View (Runtime Interaction)



6. Physical View (Deployment Diagram)

